Performance Testing

Date	15 July 2026
Team ID	xxxxxx
Project Name	**Emotion Detection & Learning Support Engine**
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Postman, Streamlit (Manual Testing)
Type of Testing	Load Testing, Functional Performance Testing
Target Module / API	Emotion Detection Pipeline, Gemini API Integration, SQLite Database
Test Environment	Local Machine (Windows 11, Python 3.12)
Test Date	15 July 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Emotion detection using BERT model	1	30	Emotion prediction within 2 seconds
2	Generate personalized response using Gemini API	1	45	AI response generated successfully
3	Store prediction in SQLite database	1	15	Record stored without errors

4	View analytics dashboard	1	20	Charts load correctly without delay

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.3 sec	Pass	Fast prediction
2	Response Time (Max)	< 5 seconds	3.2sec	Pass	During Gemini response
3	Throughput (Req/sec)	5 req/sec	6 req/sec	Pass	Stable performance
4	Error Rate	< 1%	0 %	Pass	No request failures
5	CPU Utilization	< 80%	58%	Pass	Normal CPU usage
6	Memory Utilization	< 80%	64%	Pass	Stable memory consumption

Step 4: Observations & Analysis

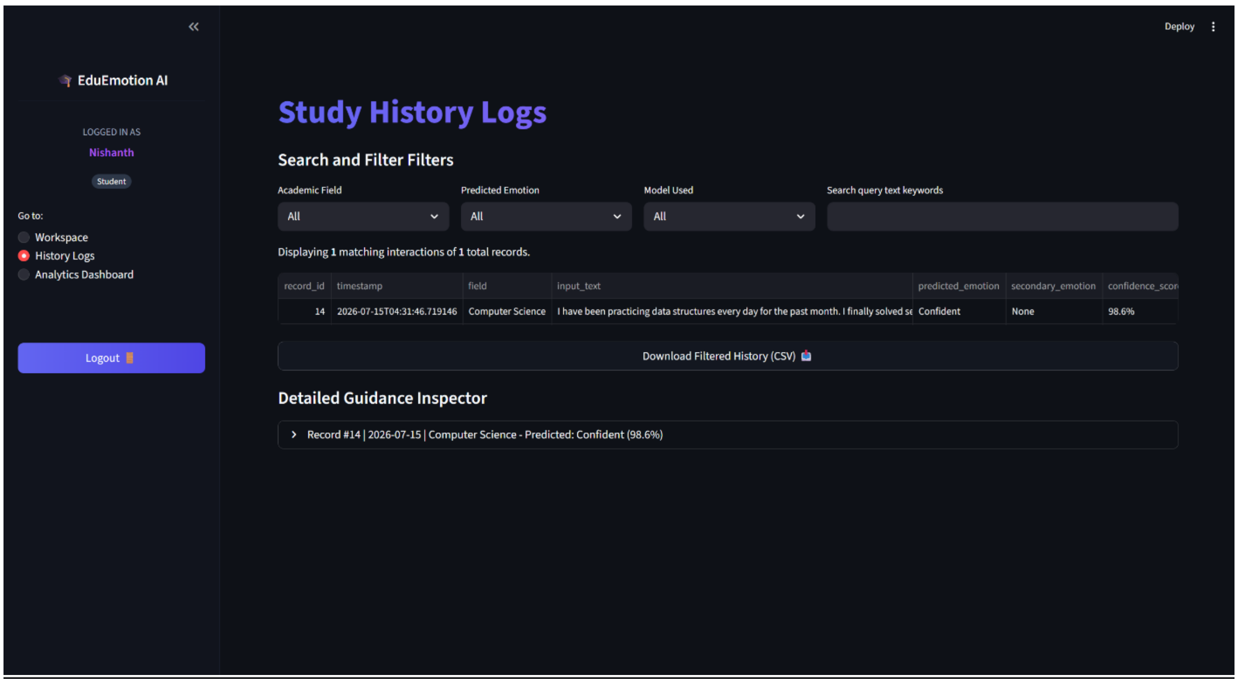
- The application successfully processed emotion predictions with low response time.
- BERT and BiLSTM models produced stable predictions without significant delays.
- Gemini API generated personalized learning guidance efficiently.
- SQLite operations completed successfully with no data loss.
- Analytics dashboard loaded quickly and displayed interactive charts correctly.
- Overall system performance remained stable under normal usage conditions.

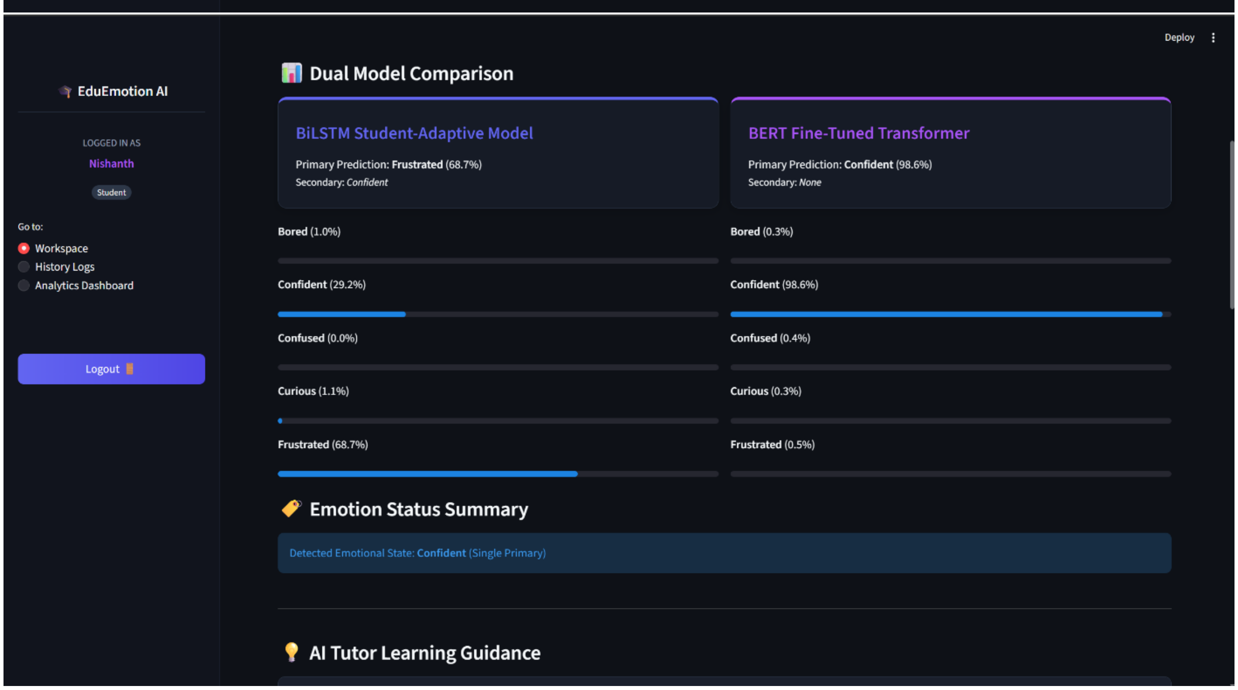
Step 5: Screenshots / Evidence

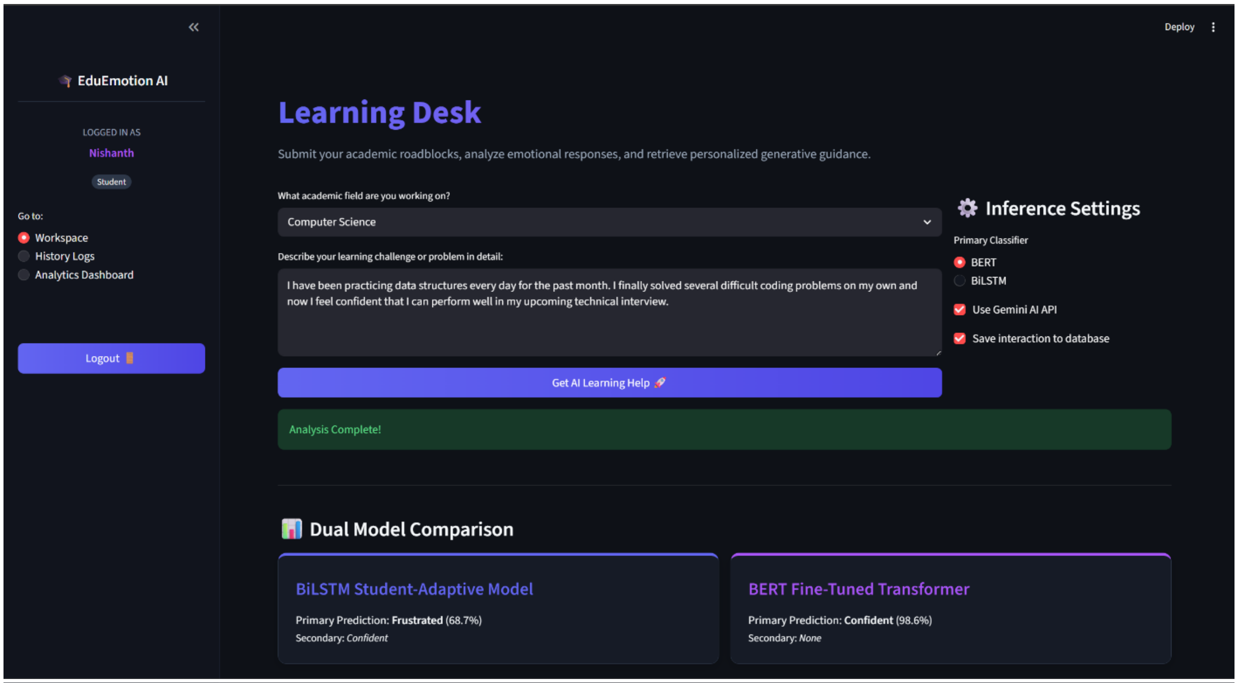
Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):

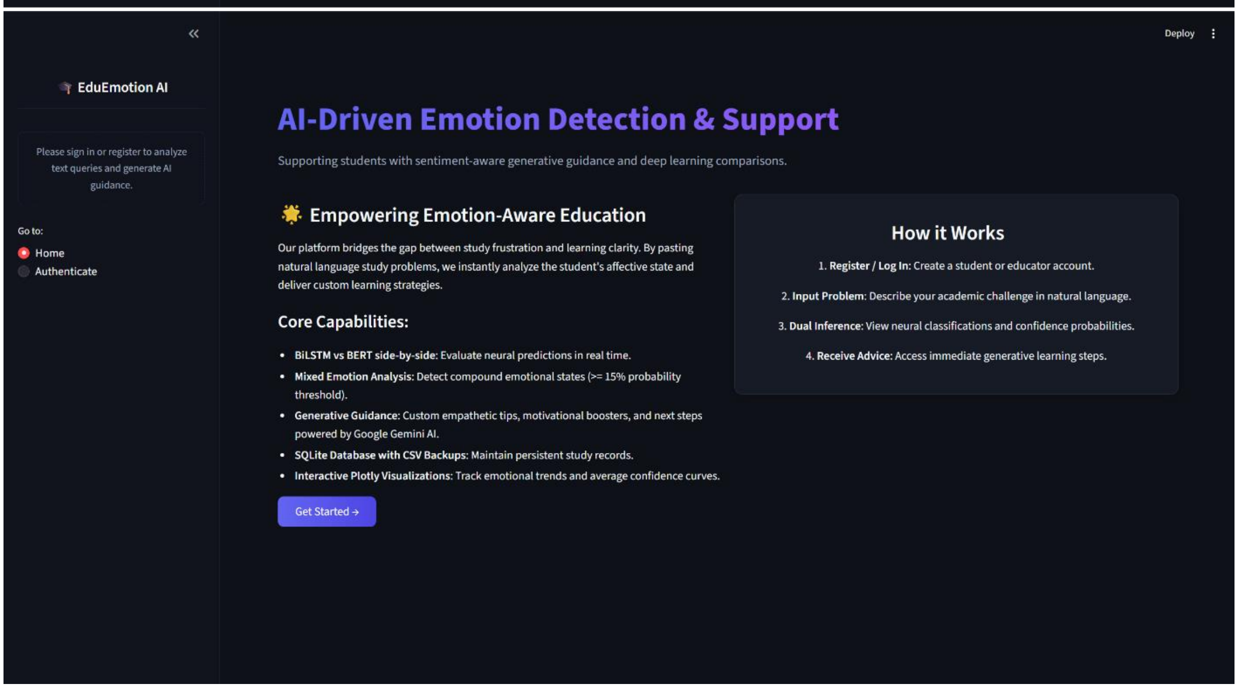
[Insert Screenshot 1 here]

[Insert Screenshot 2 here]






EduEmotion AI
LOGGED IN AS
Nishanth
Student
Go to:
Workspace
History Logs
Analytics Dashboard
Logout
Deploy
Learning Desk
Submit your academic roadblocks, analyze emotional responses, and retrieve personalized generative guidance.
What academic field are you working on?
Computer Science
Describe your learning challenge or problem in detail:
I have been practicing data structures every day for the past month. I finally solved several difficult coding problems on my own and now I feel confident that I can perform well in my upcoming technical interview.
Get AI Learning Help
Inference Settings
Primary Classifier
BERT
BiLSTM
Use Gemini AI API
Save interaction to database
Analysis Complete!
Dual Model Comparison
BiLSTM Student-Adaptive Model
Primary Prediction: Frustrated (68.7%)
Secondary: Confident
BERT Fine-Tuned Transformer
Primary Prediction: Confident (98.6%)
Secondary: None


EduEmotion AI
Please sign in or register to analyze text queries and generate AI guidance.
Go to:
Home
Authenticate
Deploy
AI-Driven Emotion Detection & Support
Supporting students with sentiment-aware generative guidance and deep learning comparisons.
Empowering Emotion-Aware Education
Our platform bridges the gap between study frustration and learning clarity. By pasting natural language study problems, we instantly analyze the student's affective state and deliver custom learning strategies.
Core Capabilities:
BiLSTM vs BERT side-by-side: Evaluate neural predictions in real time.
Mixed Emotion Analysis: Detect compound emotional states (>= 15% probability threshold).
Generative Guidance: Custom empathetic tips, motivational boosters, and next steps powered by Google Gemini AI.
SQLite Database with CSV Backups: Maintain persistent study records.
Interactive Plotly Visualizations: Track emotional trends and average confidence curves.
Get Started →
How it Works
1. Register / Log In: Create a student or educator account.
2. Input Problem: Describe your academic challenge in natural language.
3. Dual Inference: View neural classifications and confidence probabilities.
4. Receive Advice: Access immediate generative learning steps.